**Phase 3: Final Report**

# Multi-Agent Genealogical Research System with Adversarial Critique and DNA Verification

| Field | Details |
|---|---|
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| **Course** | 94815 Agentic Technologies |

## Executive Summary

This project is a working multi-agent system that investigates genealogical research questions using GEDCOM family-tree data, multi-source web retrieval (FindAGrave, Wikidata, WikiTree), CSV DNA match data, and an isolated adversarial Critic. Five LLM-driven agents and one deterministic Final Report Writer are orchestrated via LangGraph, with a Critic isolation boundary enforced as a code guarantee rather than a naming convention.

The system supports three entry modes (natural-language query, gap detection across orphaned tree branches, and subtree audit), runs against four public test trees plus the developer's personal Moore family tree, and was evaluated across an 8-case adversarial trap suite, an A/B isolation experiment, a cross-vendor Critic experiment, a 3-vendor multi-Critic ensemble experiment, and a single-agent baseline ablation.

Headline empirical results:

- **8/8 trap cases passing** across three difficulty tiers: deterministic impossibilities, plausible-but-wrong errors, and genuinely ambiguous cases.
- **Single-agent baseline 4/8 vs full pipeline 8/8.** The 4-case gap is concentrated in Tier 2 (geographic and naming plausibility) and Tier 3 (ambiguity calibration), precisely the kinds of cases the multi-agent decomposition targets.
- **Critic isolation A/B: 1/6 hypotheses changed verdict** under unfiltered Critic conditions, with measurably higher reasoning-narrative leakage in unfiltered runs.
- **Multi-Critic ensemble (Anthropic + OpenAI + Google Gemini): 2/2 ambiguous cases correctly escalated** when a single-Critic Opus 4.7 would have accepted.

Five distinct failure cases encountered during Phase 3 development are documented with root cause analysis; three at full depth, two summarized.

# Problem Framing and Agentic Justification

## The Problem

Genealogical research is the construction of family histories from incomplete, contradictory, and often unsourced records. The Genealogical Proof Standard published by the Board for Certification of Genealogists specifies that any conclusion must rest on a reasonably exhaustive search of relevant sources, complete and accurate citation, analysis and correlation of evidence, resolution of conflicting evidence, and a soundly-reasoned written conclusion. In practice, hobbyist trees published on Ancestry, FamilySearch, and similar platforms routinely violate every one of those criteria. Errors propagate widely. An unsourced parent attribution on one user's tree is frequently copied into other users' trees, and within a year that error has become "common knowledge" across the documentary record itself.

The cost of getting it wrong compounds. A single misidentified ancestor in 1850 invalidates every relationship downstream of that node, twenty to thirty descendants for a typical family tree spanning five generations. DNA matches that should support a hypothesis instead become evidence of contradiction. Subsequent researchers waste time investigating the wrong village, the wrong country, or even the wrong continent.

## Why This Is an Agentic Problem

The Genealogical Proof Standard is fundamentally a multi-stage, multi-source, adversarial reasoning task. A competent human researcher does these jobs in sequence: hunt for records, consolidate them into a candidate identity, propose relationships that fit the evidence, attack their own hypotheses for plausibility violations, repeat. Each stage uses different cognitive modes: broad search, careful disambiguation, creative inference, skeptical critique. Asking a single LLM call to do all of these at once produces output that is plausible-sounding but unreliable, because the model has no internal mechanism for noticing when its first-pass identification was wrong, when its proposed relationship violates a basic plausibility check, or when the evidence is actually genuinely ambiguous.

The empirical result of the single-agent baseline ablation (Section 5.5) makes this concrete. A single LLM call confidently accepts genuinely ambiguous cases at 0.91-0.97 confidence and silently misses geographic implausibilities (Sicily-to-Ireland in 1850) and surname-pattern violations. Multi-agent decomposition with adversarial pressure catches all of these.

## Critic Isolation

The project's core agentic claim is that **adversarial decomposition with strict information isolation between proposer and critic is what makes the critique effective**. A Critic that has access to the Hypothesizer's reasoning narrative is a Critic that anchors on the Hypothesizer's reasoning. A Critic that sees only the public hypothesis and the raw evidence has to construct its own attack independently.

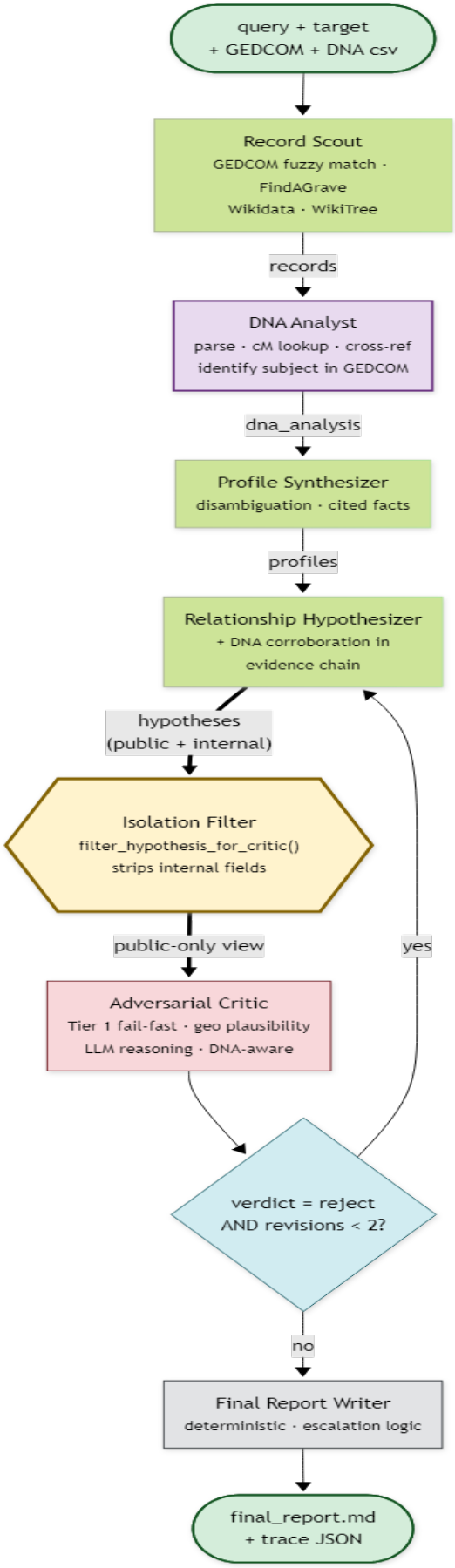
In this project, that isolation is enforced as a code guarantee: `agents/hypothesis_schema.py` declares two field sets, PUBLIC and INTERNAL. The Critic node calls `filter_hypothesis_for_critic()` exactly once at the start of its work, and never accesses raw `state["hypotheses"]` dicts at any other point.

A `state["isolation_mode"]` toggle flips the filter on and off for the A/B experiment.

The mechanical guarantee is the load-bearing architectural feature of the project.

# System Architecture

## Pipeline Overview




Six nodes orchestrated via LangGraph; five are autonomous LLM-driven agents, the sixth is a deterministic markdown composer.

1. **Record Scout** | Parses GEDCOM, extracts search criteria via LLM (including alternate name forms), runs fuzzy match against the parsed persons, and queries three external sources (FindAGrave HTML scrape, Wikidata SPARQL, WikiTree REST). Has two operational modes: query mode (default fuzzy match) and gap mode (search for plausible parents of a child with missing parental links). Retrieves but does not interpret.
1. **DNA Analyst** | Parses GEDmatch and MyHeritage match files, identifies the DNA test subject in the GEDCOM via fuzzy matching (target name + filename guess), and cross-references match names against GEDCOM persons. Predicts relationships using the Shared cM Project lookup table (DNA Painter / ISOGG ranges). Produces a `dna_analysis` state field consumed by the Hypothesizer and Critic.
1. **Profile Synthesizer** | Performs deterministic disambiguation when the Scout returns multiple candidate "subject" records (composite scoring on name similarity, birth year, and location). Consolidates the selected candidate's records into a cited profile. Every fact must reference a `record_id`. Gaps are flagged explicitly.

1. **Relationship Hypothesizer** | Proposes parent / child / sibling relationships with evidence chains, confidence scores, and stated weaknesses. Incorporates external-source corroboration (Wikidata, WikiTree, FindAGrave) into evidence chains. Appends DNA evidence items when DNA analysis identified the subject and a cross-reference exists for the proposed related person.
1. **Adversarial Critic** | Runs `filter_hypothesis_for_critic()` first, producing a public-only view of each hypothesis. Then applies, in order:
    a. **Tier 1 fail-fast**: deterministic checks (parent-younger-than-child, death-before-birth, parent-too-young-at-birth, pre-conception parental death). If any fire, the hypothesis is rejected at confidence near 0.95 without invoking the LLM at all.
    b. **Geographic plausibility:** multi-tier soft flags using haversine distance, era-aware travel feasibility, and graded verdicts (ok, flag_mild, flag_moderate, flag_strong).
    c. **DNA consistenc:** when a `dna_relevant` field is present, reasons explicitly about whether shared cM supports, contradicts, or is silent on the proposed relationship.
    d. **LLM reasoning**: full evidence-sufficiency review on the remaining hypotheses, returning verdict (accept | flag_uncertain | reject), confidence_in_critique (0..1), issues_found, evidence_cited, and justification. Self-rates `confidence_in_critique`. Production setting uses Claude Opus 4.7 (other agents use Sonnet 4.6). Maximum 2 revision cycles per hypothesis.
1. **Final Report Writer**: Deterministic, no LLM. Composes the markdown report from state and applies three-trigger escalation logic (Section 3.3). Produces the human-facing artifact.

## Orchestration Flow

```
Record Scout ──→ DNA Analyst ──→ Profile Synthesizer ──→
  Relationship Hypothesizer ⇄ Adversarial Critic (max 2 cycles) ──→
  Final Report Writer
```

The DNA Analyst was originally designed as a parallel branch off the Scout (running concurrently with the Profile Synthesizer / Hypothesizer chain) but was serialized after a bug where the Final Report Writer fired prematurely. That bug, its root cause in LangGraph's join semantics, and the resulting design lesson are documented at length in Section 6, Case 1.

## Orchestration and Memory Organization

The orchestration layer is LangGraph plus the routing function `should_revise()`. It is a workflow, not an agent. It owns four responsibilities that no individual node can see or override: state passing between nodes, conditional-edge evaluation, the revision counter that caps the Hypothesizer ⇄ Critic loop at two cycles, and the `isolation_mode` toggle that flips the Critic's view between filtered and unfiltered hypotheses for the A/B experiment. Because these guarantees live at the framework level rather than in

any agent's prompt, they cannot be silently weakened by a prompt edit. This property proved important when Section 6 Case 1 surfaced a join-semantics bug that *was* invisible to the agents themselves. The orchestrator is drawn as a wrapper around the agent nodes in the architecture diagram rather than as a node, to make the framework-vs-agent boundary visible.

Memory is organized into four tiers. **Working state** is the `GenealogyState` TypedDict, mutated in place by each agent and discarded at end of run. **Within-run trace** is `state["trace_log"]`, declared as `Annotated[list[str], operator.add]` so concurrent appends from agents that update state in the same superstep merge cleanly under LangGraph's reducer protocol. This is the only place in the system where concurrent-write coordination matters, and labelling it as a memory-organization decision rather than a plumbing detail highlights why the type annotation is load-bearing. **Episodic / persisted traces** are the JSON files in `traces/`; they are the working state of a past run captured to disk, which is what makes replay mode possible without re-invoking any LLM. **Semantic / reference memory** is the read-only domain knowledge the agents draw on: the GEDCOM file, the Shared cM Project lookup table embedded in `tools/shared_cm_lookup.py`, and the geocoding cache. There is deliberately no cross-session learned memory. Each run is stateless, which keeps the system auditable and avoids the failure mode where an early misconception poisons later runs.

One memory-control mechanism is worth calling out separately because it is the system's central agentic claim made concrete: the Critic does not have read access to the full hypotheses list. It reads through `filter_hypothesis_for_critic()`, which strips reasoning-chain fields before the Critic sees the hypothesis. This is, structurally, selective memory access where one agent's view of another agent's working state is restricted by code,.

*Figure.* `docs/screenshots/history_or_state.png` *— replay-mode sidebar with four committed demo traces, a concrete view of the episodic-memory tier.*

## Human Escalation Triggers

The Final Report Writer applies three escalation triggers post-pipeline. Flagged findings appear in a "Findings Requiring Human Review" section. Nothing is suppressed.

1. **Force-finalized after max revisions.**
   Critic rejected on both revision cycles, pipeline forced finalize. Indicates the Hypothesizer could not produce a hypothesis the Critic accepted within the budget.
2. **Low Critic confidence on accept**.
   `confidence_in_critique < 0.60` on an `accept` verdict. Indicates the Critic accepted without conviction.
3. **Conflicting verdicts across hypotheses for the same subject**.
   Accept and reject on different hypotheses about the same person. A `flag_uncertain` alongside `accept` is normal and not flagged.

The Habsburg + synthetic DNA demo trace (`traces/demos/`) exercises triggers 1 and 3 simultaneously. The Critic rejected one of two parental hypotheses on both revisions and accepted the other with high confidence.

*Figure.* `docs/screenshots/failure.png` *— Habsburg + synthetic DNA replay surfaces both triggers (force-finalize after max revisions, conflicting verdicts on the same subject) as flagged findings for human review.*

# Implementation Summary

## Tech Stack

| Layer | Choice |
|---|---|
| Orchestration | LangGraph (`langgraph`, `langchain`, `langchain-anthropic`) |
| LLMs | Claude Sonnet 4.6 (most agents); Claude Opus 4.7 (production Critic) |
| Cross-vendor experiments | OpenAI GPT-5.5 (`langchain-openai`); Google Gemini 2.5 Pro (`langchain-google-genai`) |
| GEDCOM parsing | `python-gedcom-2` |
| Fuzzy matching | `jellyfish` (Soundex, Metaphone, Jaro-Winkler, Levenshtein, composite) |
| Date handling | `python-dateutil` |
| Geocoding | `geopy` with Nominatim (rate-limited 1 req/sec) |
| External retrieval | `requests` + `beautifulsoup4` (FindAGrave); Wikidata SPARQL endpoint; WikiTree public REST |
| DNA prediction | Shared cM Project / DNA Painter / ISOGG ranges, embedded in `tools/shared_cm_lookup.py` |
| UI | Streamlit + graphviz + pandas |

API keys live in a gitignored `.env` file. `ANTHROPIC_API_KEY` is required; `OPENAI_API_KEY` and `GOOGLE_API_KEY` are optional for cross-vendor experiments.

## Three Entry Modes

The system exposes three distinct entry points, each addressing a different research workflow:

1. **Query mode** (`main.py` and the Streamlit Pipeline tab)
   Natural- language research question. Default for "who are the parents of X?" shapes. Scout fuzzy-matches the target person, full pipeline runs.
2. **Gap detection mode** (`gap_search.py` and the Streamlit Pipeline tab via mode selector)
   Scans the GEDCOM for persons missing parent links, scores plausible parents from within the tree using `tools/gap_scanner.py`, optionally runs the full pipeline on top-N broken links.
3. **Subtree audit mode** (`audit.py` and the Streamlit Audit tab)
   Given a root person and N generations, audits every parent-child relationship in the subtree. Two-pass: a deterministic pass (Tier 1 + geographic) on every relationship in seconds, followed by an optional LLM pass that runs the full pipeline on the top-N most questionable relationships from the deterministic pass.

The Streamlit app routes between modes either by an explicit mode selector or, for backwards compatibility, by keyword pattern matching on the query text.

*Figure.* `docs/screenshots/Home_Screen_Live.png` *— landing view in Live mode with the mode selector (auto-detect / Query / Audit / Gap detection) and the four-tab layout.*

## Critic Isolation Guarantee in Detail

`agents/hypothesis_schema.py` defines:

```
PUBLIC_FIELDS = {
    "hypothesis_id", "subject_id", "related_id",
    "proposed_relationship", "evidence_chain",
    "confidence_score", "stated_weaknesses",
}
INTERNAL_FIELDS = {
    "reasoning_narrative", "intermediate_steps",
    "alternatives_considered", "llm_raw_response",
}

def filter_hypothesis_for_critic(h: dict, isolation_mode: str = "filtered") -> dict:
    if isolation_mode == "unfiltered":
        return dict(h)  # used only by the A/B experiment
    return {k: v for k, v in h.items() if k in PUBLIC_FIELDS}
```

The Critic node imports `filter_hypothesis_for_critic` and applies it at the start of each critique pass. There is no other read path from the Critic to the hypothesis dict. A static check in code review catches any attempt to access `state["hypotheses"]` directly from `agents/adversarial_critic.py`.

This design is what enables the A/B isolation experiment. Toggling `state["isolation_mode"]` flips the filter at the boundary, allowing identical pipeline conditions otherwise.

*Figure.* `docs/screenshots/Evidence.png` *— JFK mother-of hypothesis card showing the full evidence chain (GEDCOM citations plus 3502 cM MyHeritage DNA evidence) and the Hypothesizer's stated weaknesses against its own claim.*

## DNA Analyst Integration into the Reasoning Loop

In an early Phase 3 build, the DNA Analyst was a "reporter." It generated a structured `dna_analysis` field that appeared in the final report's appendix but never fed back into hypothesis generation or critique. Mid-Phase 3 the DNA Analyst was reworked to:

1. Identify the DNA test subject in the GEDCOM via fuzzy matching against both the target person and the DNA filename's guessed subject.
2. Cross-reference each match name against all GEDCOM persons, surfacing per-match `gedcom_id` and `match_score` when above a 0.70 threshold.
3. Provide the Hypothesizer with a `dna_analysis` field that the Hypothesizer reads to append "DNA evidence supports..." or "DNA evidence inconsistent with..." items to the evidence chain when the subject has been DNA-tested and the proposed related person appears in the cross-reference list.
4. Provide the Critic with a `dna_relevant` field per hypothesis, containing `cm_consistency_verdict` ("supports" / "contradicts" / "no match in DNA list") which the Critic reasons about explicitly in its prompt.

This integration is dormant on data without DNA (e.g., Kennedy + no DNA file). With the synthetic DNA demos (`data/DNA_demo/`) the integration is fully exercised reproducibly without committing personal match data.

# Evaluation

## Adversarial Trap Taxonomy

Eight trap cases in `eval/trap_cases/`, manifest-driven via `eval/run_eval.py`, organized in three difficulty tiers.

**Tier 1 | Deterministic impossibilities.** Three GEDCOMs encoding rule- checkable errors: parent born after child, death before birth, parent under age 12 at child's birth. Success criterion: Critic must reject via the Tier 1 fail-fast path (no LLM step) at confidence ≥ 0.90.

**Tier 2 | Plausible but wrong.** Three GEDCOMs encoding errors that pass all Tier 1 checks but fail evidence cross-referencing. A Sicilian peasant family transposed to Dublin in 1850 (geographic implausibility); a patronymic-culture family with surname patterns that don't fit (Italian father / child surname mismatch); an 85-year father-child age gap. Success criterion: at least one hypothesis must produce verdict in {reject, flag_uncertain} at confidence ≥ 0.50, with Tier 1 NOT firing.

**Tier 3 | Genuinely ambiguous.** Two modified Kennedy GEDCOMs with near-duplicate persons in the same era. A second John Kennedy born 1917 in Boston, distinct from JFK; a second Joseph Patrick Kennedy born ~1888 in Massachusetts. Success criterion: at least one hypothesis must produce verdict `flag_uncertain` AND no hypothesis may be accept-or-reject at confidence > 0.85. This tier is not asking the system to pick the right answer; it is asking the system to acknowledge that there is no right answer.

## Trap Suite Results

```
[tier1] 3/3 passing
[tier2] 3/3 passing
[tier3] 2/2 passing
=== summary: 8/8 ===
```

All three Tier 1 cases reject at confidence ≥ 0.95 via the deterministic fail-fast (no LLM call invoked). All three Tier 2 cases produce flag_uncertain or reject with the relevant signal explicitly cited in issues_found. Both Tier 3 cases produce flag_uncertain on at least one hypothesis with no overconfident commitment to either candidate.

## Critic Isolation A/B Experiment

3 queries × 2 conditions (filtered vs unfiltered hypotheses), shared-seed methodology. Headline:

- **1/6 hypotheses changed verdict** under unfiltered conditions. The unfiltered Critic accepted a hypothesis that the filtered Critic flagged uncertain. It had been told the Hypothesizer's reasoning narrative for why this was the strongest of three candidates and anchored on that.
- **Linguistic leakage** measured via 5-gram overlap between the Hypothesizer's `reasoning_narrative` and the Critic's `justification` field nearly doubled in unfiltered runs (13 vs 7 overlapping 5-grams across 6 critiques).

Aggregate findings persisted in `eval/results/isolation_ab_summary.md`. The result establishes that the isolation enforcement mechanism does measurable work: Critics anchor on Hypothesizer reasoning when they have access to it.

The Phase 2 evaluation correctly flagged two limitations of the 5-gram leakage metric: it captures verbatim overlap only and so misses paraphrased anchoring, and the 6-hypothesis sample makes the 2:1 ratio directional rather than conclusive. Rather than improve the metric in isolation, Phase 3 triangulates the same architectural claim through two complementary methods (Section 5.4 cross-vendor Critic comparison and the three-vendor multi-Critic ensemble). If isolation didn't matter, vendor independence and ensemble aggregation would not be expected to produce different verdicts on ambiguous cases, yet they do. The leakage finding stands as a directional result. The architectural claim that isolation matters is supported collectively by the A/B experiment plus the two independent triangulations, not by any single metric.

## Cross-Vendor Experiments and Multi-Critic Ensemble

Three experiments addressed presentation feedback that same-vendor models share training data and possible failure modes.

**Cross-vendor Critic** (`eval/cross_vendor_critic_experiment.py`)
Anthropic Opus 4.7 vs OpenAI GPT-5.5 on the same hypothesis set.
On ambiguous Tier 3 cases, the OpenAI Critic produced verdicts the Anthropic Critic did not, suggesting genuinely independent failure modes.

**Asymmetric pairing** (`eval/cross_vendor_hypothesizer_experiment.py`)
Weak OpenAI Hypothesizer + strong Anthropic Opus 4.7 Critic.
Tests whether a strong adversary catches a weak proposer's mistakes.

**Multi-Critic ensemble** (`eval/multi_critic_ensemble_experiment.py`)
Anthropic Opus 4.7 + OpenAI GPT-5.5 + Google Gemini 2.5 Pro.
Conservative aggregation rule: any reject → reject; any flag_uncertain → flag_uncertain; unanimous accept required for accept. Result: **2/2 ambiguous cases correctly escalated** when single-Critic Opus would have accepted. The ensemble's escalation rate is higher than any individual Critic's rate, which is the desired direction for safety-critical genealogical claims.

## Single-Agent Baseline Ablation

Justification-of-complexity experiment. One LLM call (Sonnet 4.6, same model as the Hypothesizer), given the same fuzzy-matched GEDCOM context the multi-agent pipeline gets, asked directly for a parental hypothesis with confidence + evidence + flagged issues. No Synthesizer, no Critic, no revision loop, no Tier 1 fail-fast layer.

| Tier | Cases | Full pipeline | Baseline | Gap |
|---|---|---|---|---|
| Tier 1 (deterministic) | 3 | 3/3 | 3/3 | 0 |
| Tier 2 (plausible-wrong) | 3 | 3/3 | 1/3 | +2 |
| Tier 3 (ambiguous) | 2 | 2/2 | 0/2 | +2 |
| **Total** | **8** | **8/8** | **4/8** | **+4** |

The 4-case gap concentrates in Tier 2 and Tier 3, precisely the kinds of cases the multi-agent decomposition's specialized layers (geographic plausibility checks, adversarial Critic) were built to handle.

The Tier 1 tie is itself a useful finding: a single Sonnet 4.6 call identifies parent-younger-than-child, death-before-birth, and parent-too-young correctly when the data is laid out in front of it, rejecting at confidence 0.05 ("I'm sure this is wrong"). The full pipeline still wins on operational grounds. The deterministic Tier 1 layer fires

before the LLM is invoked, with confidence near 0.95, at zero token cost, but the value of the fail-fast layer is fail-safety and cost, not unique detection capability. This nuance is worth preserving, rather than overstating "multi-agent always wins."

The Tier 3 result is the cleanest demonstration of the value of adversarial pressure. The single-agent baseline accepted both ambiguous- Kennedy cases at 0.91 and 0.97 confidence, picking the most-plausible single answer rather than surfacing the ambiguity. The full pipeline returns flag_uncertain on both. A single LLM optimizing for a confident answer will default to one even when the evidence supports multiple interpretations; an adversarial Critic told to look for evidence- sufficiency problems finds them.

## Empirical Pipeline Runs on Real Data

In addition to the synthetic trap suite, the pipeline was tested end-to-end on:

- **Kennedy** (70 persons, English)
  JFK parental query. Target: Joseph Kennedy + Rose Fitzgerald. Correctly identified; 19th-century namesake (John Kennedy 1854) correctly disambiguated.
- **Queen** (4,683 persons, English)
  Queen Victoria parental query. Target: Edward Augustus Hanover + Victoria Mary Louisa. Correct.
- **Habsburg** (34,020 persons, German, Latin-1)
  Philip II of Spain parental query. Target: Karl V + Isabella von Portugal. Correct; English query matched German-language records via the Scout's alt-names extraction.
- **Moore Family Tree** (8,759 persons, my family's personal tree — gitignored) multiple queries against verifiable ancestors. Three-way "James Moore" disambiguation worked correctly. Trans-Atlantic geographic flag on Margaret McCusker correctly fired for known Irish ancestry.

Six pipeline traces and three audit demos are committed under `traces/demos/`. The query-mode + synthetic-DNA traces (JFK, Maria Theresia of Austria, Queen Victoria) exercise the standard pipeline end-to-end; the Maria Theresia run also triggers escalation (Critic disagreed with itself across hypotheses for the same subject, force-finalizing after two revision cycles). The gap-mode traces (Kennedy, Habsburg, Queen) demonstrate the gap-detection entry path on the same public trees; the Kennedy gap demo also exercises escalation (Critic rejected on both revisions). The audit demos (Habsburg, Queen, Kennedy subtrees) carry Pass 1 deterministic results plus Pass 2 LLM verdicts. One redacted trace from the personal Moore-family tree is committed under `traces/redacted/` demonstrating real-tree behavior with names pseudonymized via `tools/redact_trace.py`.

*Figure.* `docs/screenshots/Pipeline.png` *— the JFK trace replayed in the Pipeline tab; both parental hypotheses accept, no human review required.*

*Figure.* `docs/screenshots/AuditResults.png` *— Habsburg Maria Theresia subtree audit: 166 relationships checked deterministically, 5 flagged for Pass-2 LLM review, 161 ok.*

The Family Tree tab visualizes each saved trace's parental relationships with three distinct visual states: solid borders for relationships present in the source GEDCOM, dashed borders for gap-mode proposals (the system's filled-in candidate parent), and dashed-hollow placeholders for parental slots that remain unfilled even after gap analysis. All three states share the same Critic-verdict color palette (green for accept, amber for flag_uncertain or low-confidence accept, red for reject or escalation, gray for no verdict), so the rendering scales uniformly across query-mode, gap-mode, and replayed traces.

*Figure.* `docs/screenshots/FamilyTree.png` *— JFK family tree visualization with the three-state border legend (solid / dashed / dashed-hollow) and verdict-colored nodes; accepted parents in green at confidence 0.95 and 0.92.*

## Failure Cases

Three deep cases below; two summarized in the appendix. Each follows the form symptom → detection → root cause → fix → architectural lesson. Companion file `docs/failure_cases.md` contains the same content and is referenced from the implementation source.

## Case 1 | Final Report Writer Fired before Critic Loop Completed

**Symptom.** In an early Phase 3 build with the DNA Analyst as a parallel branch off the Record Scout, the same JFK query produced a complete report with critiques on two of three runs and an 8 KB report with empty critiques on the third. No error, no clear pattern.

**Detection.** Caught by running JFK three times back-to-back before a presentation. Diffing the trace JSONs showed the Critic node had not been entered at all in the short run.

**Root cause.** LangGraph cannot enforce deterministic join semantics when one of the incoming edges to a join node is conditional. The Final Report Writer had two incoming edges: an unconditional one from the parallel DNA Analyst branch and a conditional one from the Critic ("finalize" path). LangGraph fires a node as soon as any active path to it is satisfied; it does not wait for the conditional path to resolve. The DNA Analyst finished in ~200 ms; the Hypothesizer + Critic loop took 30-90 seconds. Whenever DNA finished first, the Writer fired immediately on the DNA path and consumed pre-Critic state.

**Fix.** Serialized the DNA Analyst between the Scout and the Profile Synthesizer. Cost: ~200 ms, dwarfed by every adjacent LLM call. The DNA Analyst's `dna_analysis` is now read directly by the Hypothesizer (for evidence-chain corroboration) and the Critic (for cM consistency checking).

**Architectural lesson.** Parallelism in agent orchestration only works cleanly when join semantics are deterministic. A conditional edge anywhere upstream of a join forces the rest of the graph downstream of the conditional to be sequential relative to that conditional path, regardless of what other paths look like. Generalizing: in any DAG-based agent framework with conditional edges, treat parallel branches that

converge at a join as sequential whenever any branch passes through a conditional. Design for the framework's failure mode (race-on-readiness) rather than the diagram's intent (wait-for-all).

## Case 2 | Cross-Surname False Positive in Fuzzy Matching at Scale

**Symptom.** A subtree audit of the Moore family tree (8,759 persons) flagged a "Joan Knorr" as a candidate mother for an ancestor whose actual mother was known and present in the tree. The flag carried high enough confidence to appear in the report's body rather than the appendix.

**Detection.** Domain-knowledge review by the developer. Joan Knorr is unrelated to the subject and her surname does not appear elsewhere in the lineage.

**Root cause.** `name_match_score()` weighted given-name and surname tokens equally and combined them into a composite score using max-of-component across Jaro-Winkler, Levenshtein, Soundex, and Metaphone. In a 70-person test tree, the only candidates above threshold tend to be genuinely related by chance. In a 8,759-person tree, that population is non-empty by chance. A given-name token match plus rough date plausibility was sufficient to push an unrelated candidate above threshold; the surname mismatch was not a hard constraint, only a component of the composite.

**Fix.** Added a surname gate at a 0.60 floor: the surname token's independent score must reach 0.60 before composite scoring runs. If the floor is not met, the candidate is dropped. Restored correct behavior on the Moore tree without regressing on the smaller trees. The 0.60 threshold was chosen empirically—high enough to reject unrelated surnames including ones that share a phoneme (Knorr vs. Moore Soundex share is below 0.60), low enough to accept common spelling variants and OCR-corrupted forms (Smyth↔Smith, Kennedy↔Kennedi).

**Architectural lesson.** Fuzzy matching at scale is a precision problem, not a recall problem. P(false positive | naive composite metric) grows roughly linearly with the candidate pool because the pool of candidates that match on a single component grows linearly while the pool that match on multiple correlated components stays small. The fix is to add gates that exploit domain structure: in genealogy, surname inheritance is a strong patrilineal signal that a naive composite blurs away. Surname spellings do drift, but cross-surname false positives at scale are far more common than the same person under wholly different surnames. The deeper takeaway is that "good enough on the small test set" is insufficient validation for systems that will run against larger production data. Phase 2 evaluation used trees of 70-4,683 persons; the bug only manifested on 8,759. Routinely running on the largest available tree as a pre-flight check would have caught it earlier.

## Case 3 | Silent Verdict Inversion from Inter-Agent Vocabulary Drift

**Symptom.** While inspecting the JFK + synthetic DNA demo trace late in Phase 3, the Critic's structured output contained `verdict: accept`, `confidence_in_critique: 0.95`, and `dna_relevant.cm_consistency_verdict: "contradicts"` simultaneously. The Critic accepted the hypothesis while its own DNA-aware logic said the DNA evidence contradicted it. The Critic's free-text justification was internally consistent; only the structured field was wrong.

**Detection.** Direct trace inspection during synthetic DNA demo work. The accept verdict on JFK's parents looked correct; the "contradicts" consistency verdict on a 3487 cM shared-DNA value (which sits exactly in the Parent/Child range of the Shared cM Project) was the contradiction that triggered closer inspection.

**Root cause.** The Hypothesizer outputs `proposed_relationship` as natural-language strings: `"father of"`, `"mother of"`. The Critic's DNA-aware logic calls `tools.shared_cm_lookup.is_consistent(claimed_relationship, shared_cM)`, which canonicalizes the relationship string against an alias table. The alias table contained `"parent"`, `"father"`, `"mother"`, `"child"`, `"son"`, `"daughter"`, but not the prepositional forms `"father of"` or `"mother of"`. When called with `"father of"`, `is_consistent()` returned `consistent=False` with deviation text `"unknown relationship type: 'father of'"`. The calling code interpreted `consistent==False` as "DNA contradicts hypothesis" without checking the deviation field.

The deeper reason this survived was that the trace recorded everything correctly. The full alias-miss was visible; the `cm_consistency_verdict: "contradicts"` was visible; the LLM's text justification reasoned correctly about the cM number itself. The structured fields and the text field disagreed, and no part of the pipeline checked them against each other.

**Fix.** Added 30+ relationship-string aliases to `tools/shared_cm_lookup.py::_CANONICAL_ALIASES`, covering the prepositional forms and gendered grand- / aunt- / uncle- / sibling-of forms. After this change, the alias table covers every relationship label the Hypothesizer is currently prompted to emit. A runtime contract assert at the boundary was flagged as future work.

**Architectural lesson.** Inter-agent vocabulary is an undocumented coupling surface. Two agents designed and tested independently agree on roughly the same relationship labels, but each agent's working set drifts over time as prompts evolve. The drift is silent because no edge of the graph type-checks the strings flowing across it.

Three structural responses are possible, in increasing cost:

- **Alias tables**
  cheapest fix (what I did). Works as long as someone remembers to update the table. Doesn't prevent the next drift.
- **Runtime contracts at boundaries**
  assert the relationship string is in a known canonical set at the moment it crosses agent boundaries. Catches drift the next time it happens.

- **Typed enum at the producer's output boundary**
  Hypothesizer emits a `RelationshipKind` enum; human-readable labels generated downstream for prose. Eliminates the drift class entirely.

The chosen alias-table fix unblocks immediate work; the typed-enum approach is future work. The generalizable lesson: any structured field that flows from one agent into another's structured logic should be treated like a typed API, even when expressed as a string. Free-text inter-agent fields should never feed structured downstream logic without canonicalization at the boundary.

This is also a case where the trace artifact saved me. The bug had shipped for at least a week before being noticed because the LLM-text justifications looked plausible end-to-end. Routine inspection of structured fields against each other for internal consistency is now a review-time check.

## Additional Narrower Failures

`python-gedcom-2` **Latin-1 GEDCOM crash.** The library's `Parser.parse_file()` hardcodes `line.decode('utf-8-sig')` and provides no encoding override. The Habsburg GEDCOM is Latin-1 (German umlauts). First header line containing `Österreich` raised `UnicodeDecodeError`. Worked around by reading the file ourselves with `encoding='latin-1', errors='replace'` and calling `parse_gedcom_text()` on the resulting string. One-line workaround at every call site.

**Gap-mode produced zero hypotheses.** In an early gap-detection build, the Scout's gap mode wrote candidate parent records to `state["gap_candidates"]` while the Hypothesizer read from `state["retrieved_records"]`. Pipeline ran without error, produced zero hypotheses on every gap query. Detected on the Moore tree (1,875 gaps in the summary table, no Hypothesizer output). Fixed by normalizing the Scout's gap-mode output to write to the standard `retrieved_records` state key, so the Hypothesizer reads from a single source of truth regardless of mode. Lesson: state-key contracts between agents need to be tested with a passing end-to-end run, not just unit tests on each agent.

# Risk and Governance

## Privacy and PII

Genealogical data is permanent. A person cannot change their family tree or their DNA in response to a leak. A real DNA match list is identification-grade. Match names, GUIDs, and shared cM values are sufficient to identify individuals across platforms and re-identify anonymous DNA datasets. The project handles this through several layered controls:

1. **Personal trees gitignored.**
   `data/PII Trees/` and `data/DNA/` never enter git history. Verified via `git check-ignore`.

2. **Synthetic demo files committed.**
   `data/DNA_demo/` contains synthetic MyHeritage CSV files matching public Kennedy / Habsburg / Queen GEDCOM persons. Names are deliberately public-figure names from the public GEDCOMs; cM values are realistic Shared cM Project distributions. Reproducible without committing personal match data.
3. **Trace redactor**
   `tools/redact_trace.py` pseudonymizes names → `PERSON_NNN` and DNA GUIDs → `MATCH_NNN` while preserving cM values, dates, relationships, and reasoning structure. Produces a sibling `.map.json` mapping table that stays gitignored. `traces/redacted/` is allow-listed in the gitignore for committable artifacts; `traces/redacted/*.map.json` is denied.

## Potential Commercial-Grade Controls

For a hypothetical productionized version, the project's documentation section discusses standard commercial controls:

- **Tokenization before LLM.** A deterministic pre-processor replaces names, addresses, DOBs, and DNA GUIDs with opaque tokens before prompts are built. The LLM reasons over tokens; a post-processor rehydrates tokens to real values in the final output. Microsoft Presidio and AWS Comprehend PII implement this pattern.
- **Vendor data-processing agreements.** Anthropic, OpenAI, and Google enterprise-tier API contracts include DPAs that bind vendors to GDPR/CCPA terms, prohibit training on inputs by default, and define breach notification. HIPAA Business Associate Agreements are required for health data and are available on enterprise tiers.
- **No-training defaults.** Paid API tiers from all three vendors do not train on inputs. Free chat tiers (chatgpt.com, claude.ai free) often do retain conversations.
- **Encryption + access control.** TLS 1.2+ in transit, KMS-managed keys at rest, RBAC on log access, stricter access on traces (which capture full prompts) than on source data.
- **Audit logging + data minimization.** Log who queried what when, but log redacted prompts. Retain on a rolling window with automated purge.
- **Right-to-deletion.** GDPR Article 17 / CCPA require findable + deletable artifacts. Easier with database-backed storage keyed by subject ID than with scattered files.
- **Local LLMs for highest sensitivity.** Run a local Llama-class model (Ollama, vLLM) so PII never leaves the user's machine. Gold standard for genealogy + DNA. Quality trade-off, but the threat model is asymmetric. DNA data can't be revoked.

## Misuse Risks

The system can construct plausible-sounding family histories for adversarial users. Mitigations:

- **Adversarial Critic + escalation.**
  The Critic's job is to attack the Hypothesizer's claims. Escalation triggers (low Critic confidence, conflicting verdicts, force-finalized) surface findings that should not be acted on without human review.
- **Sourcing requirement.**
  Every fact in a profile must cite a `record_id`. Hypotheses without sources are flagged.
- **Confidence calibration.**
  Tier 3 evaluation explicitly measures whether the system commits confidently to ambiguous answers. Current result: 2/2 ambiguous cases produce flag_uncertain rather than confident accept/reject.

## Reliability

LLM calls are non-deterministic. Mitigations:

- **Deterministic checks first.**
  Tier 1 fail-fast and Profile Synthesizer disambiguation are deterministic. LLM calls run only on cases that survive the deterministic layer.
- **Trace artifacts.**
  Every run writes `trace_<timestamp>.json` and `.md`. Reproducibility is best-effort (LLMs change), but the JSON records the full state, every prompt, every response, and every agent's trace_log. Bugs are diagnosable post-hoc.
- **Multi-Critic ensemble**
  Available as a higher-cost option for safety- critical claims.

## Reproducibility without API Access

A grader requirement stated at project outset was the ability to evaluate the system end-to-end without configuring an Anthropic API key. The system supports this through a deterministic replay mode in two surfaces:

- **CLI replay**
  `python main.py --replay <trace.json>` loads a saved trace and prints the agent trace log, the summary panel, and the final report; no LLM calls.
- **Streamlit replay**
  A sidebar Mode toggle (Live | Replay) switches the Pipeline tab to a trace-picker. Family Tree and DNA Analysis tabs render automatically from the loaded trace; the Audit tab has a separate loader for saved audit JSONs (Pass 1 + Pass 2 results).

Six pipeline-trace demos (three query + DNA, three gap-mode) and three subtree-audit demos cover all three entry modes plus the DNA and escalation paths against

the public-data trees (Kennedy, Habsburg, Queen). One redacted trace from the personal Moore tree demonstrates real-data behavior with names pseudonymized via `tools/redact_trace.py`.

The replay artifacts are intentionally opaque to the LLM API: a grader who clones the repo, installs Python dependencies, and never configures an API key can still walk the four-tab Streamlit interface end-to-end and reach every architectural feature documented in this report.

*Figure.* `docs/screenshots/Options.png` *— the mode selector and the sidebar Live / Replay toggle in one frame; the Replay path is the API-key-free reproducibility route called out above.*

## Future Work

In rough priority order:

**Migration map visualization**
This is a passion project feature that became quite a sidequest over the course of this project, so it earns its place as top priority by already being partially developed and of most personal interest to me. It is a departure from the adversarial critique and research support focus of this project overall, though.

The general idea is to, using GEDCOM files as input, generate and dynamic map view over time that shows the family tree and underlying maps evolve together. The scoped approach, includes time-stepped historical boundaries, lineage narrative mode with `fly_to` camera transitions, period-correct cartographic style, and DNA admixture probability clouds. Conceptual design in earlier scoping documents; PoC code in `app/` was not integrated with pipeline output.

**DNA clustering**

DBSCAN or hierarchical clustering on shared-match graphs. The DNA Analyst currently uses point-wise lookup. Clustering could reveal shared-match groups that suggest a common ancestor not in either match's tree.

**Typed inter-agent contracts**
Convert relationship strings (and similar drift-prone vocabularies) to typed enums at producer boundaries. Closes the failure mode in Section 6, Case 3.

**Issue-aware revision routing**
The current Critic→Hypothesizer revision loop is uniform. Any rejection routes back to the same agent regardless of the failure type. The more sophisticated design I hope to implement will inspect the Critic's `issues_found` and route to the agent best positioned to address each (Scout for insufficient retrieval, Synthesizer for wrong-candidate disambiguation, Hypothesizer for weak reasoning). Trade-offs that need empirical evaluation before committing:

- Critic classification accuracy becomes a new bottleneck.
- Cycle/ping-pong termination must be enforced at the graph level.
- Per-agent revision bounds replace the single global counter.
- The isolation A/B experiment should be re-run since classification adds a new information surface in the Critic's output

With the increased development burden and limited access to some of the external tools that would best support the Scout in actually delivering value for thisfeature, the rejection routing was excluded from project scope.

**Active learning for Critic threshold tuning**
Findings flagged by human review would become training data for a thin classifier that adjusts the Critic's confidence floor on similar future cases.

**Handwritten record OCR**
Supporting documents are often digitized, but as images of handwritten text. Such primary documentary evidence (parish registers, census images) currently cannot enter the pipeline but would be valuable additions.

**FamilySearch API integration.**
Stubbed in `tools/familysearch_search.py`; requires API key registration, available only to product developers. Wikidata + WikiTree provide independent sources without the registration cost.

**Multi-Critic ensemble in production.**
Implemented as merely an experiment here after presentation feedback, the results suggest this could be a configurable production option for high-stakes queries.

## Reflection

The project's biggest takeaway for me was the value a multi-agent structure provides as compared to a single-agent. The reason the multi-agent decomposition matters is not that it catches things a single LLM *can* not, but that it catches things a single LLM *does* not. Without an explicit Critic prompted to look for geographic plausibility violations, the LLM does not check for them. Without an explicit Critic told to surface ambiguity, the LLM commits to a confident answer. The agentic value is in the prompted decomposition and the adversarial pressure, not in any individual agent's superhuman reasoning.

The project's hardest design decision, in retrospect, was the Critic isolation enforcement mechanism. The A/B experiment was only possible because the filter was a single function, and the lesson from Section 6 Case 3 (silent verdict inversion) was only catchable because the structured field flowed across a known boundary that I could inspect. Naming-convention isolation would have eroded under prompt evolution. Here, the function-as-boundary held.

The project's biggest unexpected lesson came from the parallel-branch report writer bug (Section 6, Case 1). The framework's actual semantics do not match the framework's diagrammatic intent when conditional edges are present. Learning to design for the framework's failure mode rather than the diagram's promise generalized to subsequent design decisions elsewhere in the project.

Things I would do differently:

- The trace_writer was added late in Phase 2, replacing an ad-hoc print-based logging scheme. Adding it earlier would have caught Case 3 sooner.

- Routinely running the full pipeline on the largest available tree as a pre-flight check would have caught Case 2 sooner.
- Documenting inter-agent vocabularies in a single canonicalization module from the start would have prevented Case 3 entirely.

All three are corollaries of the same broader observation: The artifacts a multi-agent system produces (traces, logs, structured fields) are not just evaluation outputs; they are diagnostic surfaces that should be inspected as a routine part of development, not just at the end.

Things I am proud of:

- The Critic isolation enforcement mechanism
- the escalation logic on the Final Report Writer
- the synthetic DNA demos that make the DNA path reproducible without exposing personal data
- the trace redactor that makes personal-tree work shareable in redacted form.

Each of those was a small piece of code that reduced a class of failures rather than catching one specific instance. They also helped me carry something from the personal, tinkering realm into a more public and academic forum.

The project began as an exercise in agentic decomposition for a genealogical research workflow. It ended having produced empirical evidence that adversarial decomposition with strict isolation catches specific classes of error that monolithic prompting cannot.